Digital Asset Cryptography

Part 2 - Bitcoin: Exposure & Migration Analytics

Compiled by the Spaced Research Ledger

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Somewhere between 1.6 million and 7 million bitcoin are exposed to a future quantum attacker, depending on who is counting and what they count. That spread is the central finding of this part, and it is a definitional dispute rather than a measurement one. The analytics firms broadly agree on the underlying blockchain data. They disagree about whether coins that owners could protect today belong in the same total as coins that cannot be protected at all. No headline exposure figure from this subject means anything without its definition attached.

Section 1 explains what makes a coin exposed and the three tiers that exposure sorts into. Section 2 covers the trackers publishing live totals. Section 3 covers the lower estimates and the methodological fight over them. Section 4 covers dormant coins, Satoshi Nakamoto's holdings, and what the movement data actually shows.

1. What Counts as Exposed

A bitcoin is quantum-vulnerable only when its public key is visible on-chain. Address types differ on when that happens, and the difference sorts the exposure into tiers.

Current State

Exposure sorts into three tiers by when the public key becomes visible.[1] Tier A is visible immediately and is the long-range target. It covers P2PK, an early format that publishes the key directly and holds about 1.7 million BTC, including Satoshi Nakamoto's estimated 1.1 million across roughly 22,000 addresses. It also covers bare multisig and Taproot, whose tweaked public key is on-chain by construction.[1]

Tier B is hidden until spent, and therefore preventable. The common modern formats stay safe while unused, but become exposed through address reuse after spending, through spending on forked chains, and through shared extended public keys. About 5.2 million BTC is estimated at risk this way.[1] Tier C is the short-range window: every address type exposes its public key while a transaction waits unconfirmed in the mempool.[1]

The two attack classes need very different hardware. A long-range attack on stored coins works if a key can be broken over days or weeks. Short-range hijacking of an in-flight transaction means breaking a key in minutes, a substantially later capability.[1] On this tiering, roughly 35% of BTC sits in theoretically vulnerable addresses, and about 65% sits in address types whose keys have never been revealed.[1]

2. The Trackers and Their Numbers

Three organisations publish running totals of exposed bitcoin. Their headline figures agree within about 15%, which is closer than the public debate suggests.

Recent Developments

The August 2026 readings cluster near seven million coins. Project Eleven's Bitcoin Risq List reported 7,039,051 BTC at risk as of 20 July 2026.[2] ChainQuery reported 7,052,314 BTC, about 35.1% of circulating supply, as of 2 August 2026, split into 1,934,419 always-exposed and 5,117,895 reuse-exposed coins.[3] Glassnode Research estimated 6.04 million BTC, or 30.2% of supply, exposed at rest as of August 2026, split into 1.92 million structurally exposed and 4.12 million operationally exposed.[4]

Current State

The Risq List is an open-source record of public-key exposure at the script and address-reuse level, and the closest thing to a canonical real-time public count.[5] It puts approximately 6.9 million BTC, roughly one third of circulating supply, in quantum-vulnerable addresses.[5] The full report behind it, published 6 May 2026, runs to 110 pages.[5]

The decomposition matters more than the total. Address reuse accounts for about 4.99 million BTC, or 72.3% of the exposed figure. Legacy P2PK scripts account for about 1.72 million, or 24.8%. Taproot holds about 198,000 exposed coins, because its x-only public key is visible in the address, and bare multisig is negligible by comparison.[5]

3. The Lower Estimates and the Dispute

Not everyone counts seven million. The disagreement is about which risks belong in the number, not about the blockchain data.

Current State

Coinbase convened an Independent Advisory Board on Quantum Computing and Blockchain, with professors from UT Austin, Stanford and UCSB alongside Ethereum Foundation and Coinbase representatives. In a June 2026 document the board counts approximately 1.7 million Bitcoin across about 20,000 public keys in the directly exposed P2PK form.[6] The same coverage attributes about 5 million BTC of address-reuse exposure to Project Eleven, consistent with the Risq List's own figure.[6] The two organisations agree on the underlying counts.[6]

Galaxy Digital's March 2026 report concludes the vulnerability is significant but manageable. It puts roughly 7 million BTC, about $470 billion at then-recent prices, at risk under a broad long-exposure definition.[7] It states plainly that other estimates come in lower depending on methodology.[7]

CoinShares counts differently. Its February 2026 research treats only the roughly 1.6 million BTC in legacy P2PK addresses, about 8% of supply, as theoretically exposed. It then narrows the practically significant figure much further, to 10,200 coins sitting in holdings whose theft could cause any appreciable market disruption.[8]

The narrowing argument is distributional. The vulnerable coins spread across roughly 32,607 individual holdings of about 50 BTC each, which CoinShares argues would take millennia to crack even under outlandishly optimistic scenarios.[8] Its earlier research found that almost all vulnerable coins are mining rewards that have never moved.[9] It counted 34,068 P2PK addresses holding between 49 and 51 BTC, and concluded that only about 24 addresses warrant serious concern.[9]

A widely repeated middle figure also circulates: 6.7 million BTC, about $600 billion, vulnerable through unsafe address type or address reuse. Up to 2.3 million of that is considered dormant, and dormancy is framed as making it an especially attractive target.[10]

Unresolved Conflicts

The main dispute is definitional. CoinShares writes that claims of 25% vulnerability often include temporary risks, such as reused exchange addresses, which can be easily mitigated.[8] The Risq List counts exactly those reused addresses and reaches roughly one third of supply.[5] Both sides read the same blockchain. What would settle it is a shared convention on whether remediable exposure belongs in the headline number, and no such convention exists.

A smaller discrepancy sits inside the agreeing camp. The exposure-tier analysis puts the 1.7 million P2PK coins across roughly 22,000 addresses, while the Coinbase board's document says about 20,000 public keys.[1][6] The totals match and the key counts do not, a methodology difference the sources leave unexplained.

4. Dormant Coins and Migration Signals

If old coins started moving in bulk, it could mean holders were fleeing quantum risk. The data shows the opposite.

Recent Developments

In August 2026, six wallets untouched for more than ten years moved roughly $40 million, and most of it did not go to exchanges.[11] The trend direction cuts against the panic narrative. Dormant-coin activity is at its lowest level since 2022, and 2026 is on pace for under half of the previous year's total. Old coins are moving less, not more.[11]

Current State

Satoshi Nakamoto's estimated 1.1 million BTC, worth over $85 billion, has not moved since 2010. Prediction-market participants have staked over $2.6 million on the question, with odds strongly favouring continued dormancy through December 2026.[12]

Migration into quantum-resistant addresses cannot be measured yet, since Part 3 shows none have activated. The nearest proxy is Taproot's share of transactions, which peaked at about 42% in 2024, driven largely by inscription activity.[13] It settled to roughly 15% to 20% by late 2025 as that activity declined.[13]

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from three to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

Sources

1. onrampbitcoin.com — which bitcoin is vulnerable to quantum computing address types exposure tiers and what you can do - https://onrampbitcoin.com/knowledge-center/which-bitcoin-is-vulnerable-to-quantum-computing-address-types-exposure-tiers-and-what-you-can-do (undated)
2. Project Eleven - Bitcoin Risq List - FAQs - https://bitcoin-risq-list.projecteleven.com/faqs (as of 2026-07-20)
3. Bitcoin Quantum Exposure: 7.1M BTC at risk · ChainQuery.com - https://chainquery.com/reports/quantum-exposure (as of 2026-08-02)
4. StarkWare’s quantum-safe Bitcoin transaction reaches mainnet - https://www.cryptopolitan.com/starkware-quantum-bitcoin-mainnet/ (as of 2026-08-27)
5. postquantum.com — project eleven quantum threat blockchains 2026 - https://postquantum.com/industry-news/project-eleven-quantum-threat-blockchains-2026/ (as of 2026-05-06)
6. Coinbase Quantum Advisory Council Publishes Position Paper on Quantum Computing and Blockchain - https://www.coinbase.com/blog/coinbase-quantum-advisory-council-publishes-position-paper-on-quantum-computing-and-blockchain (as of 2026-06-08)
7. coindesk.com — bitcoin s quantum threat is real but far from an existential crisis galaxy says - https://www.coindesk.com/tech/2026/03/19/bitcoin-s-quantum-threat-is-real-but-far-from-an-existential-crisis-galaxy-says (as of 2026-03-19)
8. coinshares.com — quantum vulnerability in bitcoin a manageable risk - https://coinshares.com/insights/research-data/quantum-vulnerability-in-bitcoin-a-manageable-risk/ (as of 2026-02-06)
9. coinshares.com — burning quantum vulnerable coins is a bad idea - https://coinshares.com/insights/research-data/burning-quantum-vulnerable-coins-is-a-bad-idea/ (as of 2025-08-05)
10. bitcoinsuisse.com — bitcoin takes first steps toward post quantum resistance - https://bitcoinsuisse.com/industry-blog/bitcoin-takes-first-steps-toward-post-quantum-resistance (undated)
11. coindesk.com — bitcoin wallets untouched for 10 years moved usd40 million most avoided exchanges - https://www.coindesk.com/markets/2026/08/28/bitcoin-wallets-untouched-for-10-years-moved-usd40-million-most-avoided-exchanges (as of 2026-08-28)
12. financefeeds.com — will satoshi nakamoto ever move his bitcoin - https://financefeeds.com/will-satoshi-nakamoto-ever-move-his-bitcoin/ (undated)
13. spark.money — bitcoin segwit adoption tracker - https://www.spark.money/tools/bitcoin-segwit-adoption-tracker (undated)